

Product Environmental Profile

Relay actuator with control key
MyHome



BTICINO'S ENVIRONMENTAL COMMITMENTS

- Incorporate environmental management into our industrial sites**

Of all Legrand sites worldwide, over 85% are ISO 14001-certified [sites belonging to the Group for more than five years].

- Offer our customers environmentally friendly solutions**

Develop innovative solutions to help our customers design more energy efficient, better managed and more environmentally friendly installations.

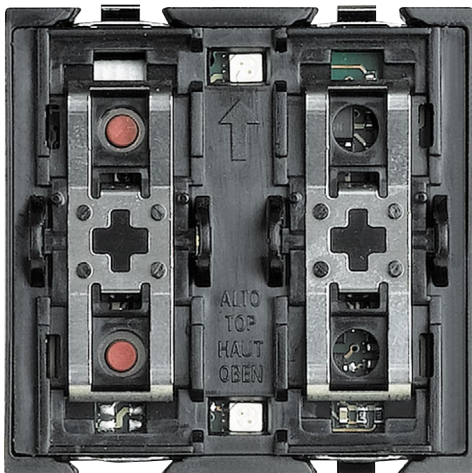
- Involve the environment in product design and provide informations in compliance with ISO 14025**

Reduce the environmental impact of products over their whole life cycle.

Provide our customers with all relevant information (composition, consumption, end of life, etc.).



REFERENCE PRODUCT

Function	To allow the command of a single load: resistive, for ferromagnetic transformers and fluorescent lamps. To be completed with two module key cover.
Reference Product	 BT-L4671/1 Relay actuator with control key - 27 V d.c. - LivingLight series - MyHome

The company reserves the right to change specifications and designs without notice. All illustrations, descriptions, dimensions and weights in the document are for guidance and cannot be held binding on the company.



PRODUCTS CONCERNED

The environmental data is representative of the following products:

BT-L4671/1

BT-H4671/1, BT-AM5851/1

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■ CONSTITUENT MATERIALS

This Reference Product contains no substances prohibited by the regulations applicable at the time of its introduction to the market. It respects the restrictions on use of hazardous substances as defined in the RoHS directive 2011/65/EU.

Total weight of Reference Product		80 g (with unit packaging)			
Plastics as % of weight		Metals as % of weight		Other as % of weight	
Polycarbonate	24,1 %	Copper alloys	0,8 %	Electronic cards	36,1 %
Polyamide	1,8 %	Steel	0,8 %	Packaging as % of weight	
				Wood	21,4 %
				Paper / cardboard	14,0 %
				Polyethylene	0,5 %
				PET	0,5 %
Total plastics	25,9 %	Total metals	1,6 %	Total other and packaging	72,5 %

Estimated recycled material content: 14 % by mass.



■ MANUFACTURE

This Reference Product comes from sites that have received ISO14001 certification.



■ DISTRIBUTION

Products are distributed from logistics centres located with a view to optimize transport efficiency. The Reference Product is therefore transported over an average distance of 780 km by road from our warehouse to the local point of distribution into the European market.

Packaging is compliant with european directive 2004/12/EU concerning packaging and packaging waste. At their end of life, its recyclability rate is 94 % (in % of packaging weight).



■ INSTALLATION

For the installation of the product, only standard tools are needed.



■ USE

Under normal conditions of use, this product requires no servicing, no maintenance or additional products.

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END OF LIFE

The product end-of-life factors are taken into account during the design phase. Dismantling and sorting of components or materials is made as easy as possible with a view to recycling or failing that, another form of reuse.

• Recyclability rate:

Calculated using the method described in technical report IEC/TR 62635, the recyclability rate of the product is estimated at 82 %. This value is based on data collected from a technological channel operating on an industrial basis. It does not pre-validate the effective use of this channel for the end of life of this product.

Separated into:

- plastic materials (excluding packaging)	: 25 %
- metal materials (excluding packaging)	: 2 %
- other materials (excluding packaging)	: 21 %
- packaging (all types of materials)	: 34 %



ENVIRONMENTAL IMPACTS

The evaluation of environmental impacts examines the stages of the Reference Product life cycle: manufacturing, distribution, installation, use and end-of-life. It is representative from products marketed and used in Europe, in compliance with the local current standards.

For each phase, the following modelling elements were taken in account:

Manufacture	Materials and components of the product, all transport for the manufacturing, the packaging and the waste generated by the manufacturing. In the analysed configuration the key cover is not been considered.
Distribution	Transport between the last Group distribution centre and an average delivery point in the sales area.
Installation	The end of life of the packaging.
Use	• Product category: active product. • Use scenario: ten-year working life. Active mode power: 0,45 W for 33 % of the time; stand-by mode power: 0,12 W for 67 % of the time. This modelling duration does not constitute a minimum durability requirement. • Energy model: Electricity Mix, Europe 27 - 2002.
End of life	The default end of life scenario maximizing the impacts.
Software and database used	EIME V5 and its database «CODDE-2015-04»

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SELECTION OF ENVIRONMENTAL IMPACTS

	Total for Life cycle		Raw material and manufacture		Distribution		Installation		Use		End of life	
Global warming	1.28E+01	kgCO ₂ eq.	1.01E+00	8%	3.11E-03	< 1%	1.60E-03	< 1%	1.18E+01	92%	5.89E-03	< 1%
Ozone depletion	3.07E-06	kgCFC-11 eq.	1.99E-07	6%	6.29E-12	< 1%	8.80E-12	< 1%	2.87E-06	94%	1.49E-10	< 1%
Acidification of soils and water	9.07E-02	kgSO ₂ eq.	1.24E-03	1%	1.40E-05	< 1%	7.42E-06	< 1%	8.94E-02	99%	2.24E-05	< 1%
Water eutrophication	3.77E-03	kg[PO ₄] ³⁻ eq.	3.89E-04	10%	3.21E-06	< 1%	4.86E-06	< 1%	3.35E-03	89%	2.58E-05	< 1%
Photochemical ozone formation	4.41E-03	kgC ₂ H ₄ eq.	1.85E-04	4%	9.92E-07	< 1%	5.28E-07	< 1%	4.23E-03	96%	1.75E-06	< 1%
Depletion of abiotic resources - elements	2.47E-04	kgSb eq.	2.46E-04	100%	1.24E-10	< 1%	6.89E-11	< 1%	5.39E-07	< 1%	3.77E-10	< 1%
Total use of primary energy	2.19E+02	MJ	1.52E+01	7%	4.16E-02	< 1%	2.11E-02	< 1%	2.04E+02	93%	6.25E-02	< 1%
Net use of fresh water	3.95E-02	m ³	8.65E-03	22%	2.78E-07	< 1%	3.72E-07	< 1%	3.08E-02	78%	5.13E-06	< 1%
Depletion of abiotic resources - fossil fuels	1.33E+02	MJ	1.12E+01	8%	4.36E-02	< 1%	2.25E-02	< 1%	1.22E+02	91%	8.39E-02	< 1%
Water pollution	7.04E+02	m ³	2.07E+02	29%	5.11E-01	< 1%	2.52E-01	< 1%	4.96E+02	70%	6.67E-01	< 1%
Air pollution	6.25E+02	m ³	1.17E+02	19%	1.27E-01	< 1%	1.31E-01	< 1%	5.07E+02	81%	6.93E-01	< 1%

The values of the 27 impacts defined in the PCR-ed3-EN-2015 04 02 are available in the digital database of pep-ecopassport.org website.

For products covered by the PEP other than the Reference Product, the environmental impacts of each phase of the lifecycle are assimilated to the impacts of the Reference Product.

Registration N°: LGRP-00332-V01.01-EN	Drafting rules: PEP-PCR-ed3-EN-2015 04 02 Supplemented by PSR-0005-ed2-2016 03 29
Verifier accreditation N°: VH02	Information and reference documents : www.pep-ecopassport.org
Date of issue: 12-2016	Validity period: 5 years
Independent verification of the declaration and data, in compliance with ISO 14025:2010 Internal ☒ External ☐	
The PCR review was conducted by a panel of experts chaired by Philippe Osset (SOLINNEN)	
The elements of the present PEP cannot be compared with elements from another program	
Document in compliance with ISO 14025 : 2010: «Environmental labels and declarations. Type III environmental declarations»	
Environmental data in alignment with EN 15804 : 2012 + A1 : 2013	

